A065 v0.3.5 | High-resolution historical trefoil falsifier

Unique run: A065-v0.3.5__20261018T230012125250Z-bb43eeb834

Command: HIGHRES; native status: EXECUTED

1. Scientific contract

Independent precommitted geometric seed arrays, numerical reduced regularized filament ODE. Original D01/D02/D03 source seals untouched.

Preregistration SHA256: 2e36f33e328020fd0942d89573fda8a774f6e41e1e2a8245ef46edf33f23113a

N=600, 1200, 2400, 4800. gamma=1, T=0.04; RK4 8 steps; epsilon=0.06/0.12/0.24. Original dimensionless geometry unscaled.

Every mesh is direct analytic arclength sampling; no new independent physical fluid provider.

Spatial E(N,2N)=RMS(final(N)-final(2N)[::2]) / RMS(final(N)-initial(N)).

Registered baseline acceptance eps=0.12: spatial E < 1% for all pairs and both trefoils.

0.1% is an aspirational reporting target, not a replacement threshold.

2. Gates and epistemic limits

G15 input/hash PASS on successful pre-run validation. G16 baseline spatial convergence.

G17 all 3 epsilon diagnostic scans. G18 time and limited Python/native parity.

G2 legacy FAIL, G13 historical coarse-resolution FAIL retained; Fremlin G11 and physical G3 are independent.

Euler PDE, PKLSA time dynamics, Casimir, pressure-wall response, gravity and experimental calibration: UNRESOLVED.

3. Executed quantitative results

A_Fourier_2015_p32

Baseline 1%: PASS; temporal PASS; CPU Python/native N150 PASS

eps=0.06: 600_vs_1200 | relative=0.849902% | abs_RMS=0.00013543366 | 1%=PASS | 0.1% target=FAIL

eps=0.06: 1200_vs_2400 | relative=0.211758% | abs_RMS=3.3919687e-05 | 1%=PASS | 0.1% target=FAIL

eps=0.06: 2400_vs_4800 | relative=0.052870% | abs_RMS=8.4798691e-06 | 1%=PASS | 0.1% target=PASS

eps=0.12: 600_vs_1200 | relative=0.228330% | abs_RMS=3.346035e-05 | 1%=PASS | 0.1% target=FAIL

eps=0.12: 1200_vs_2400 | relative=0.057009% | abs_RMS=8.3648808e-06 | 1%=PASS | 0.1% target=PASS

eps=0.12: 2400_vs_4800 | relative=0.014248% | abs_RMS=2.0912065e-06 | 1%=PASS | 0.1% target=PASS

eps=0.24: 600_vs_1200 | relative=0.064356% | abs_RMS=8.0676555e-06 | 1%=PASS | 0.1% target=PASS

eps=0.24: 1200_vs_2400 | relative=0.016083% | abs_RMS=2.0168673e-06 | 1%=PASS | 0.1% target=PASS

eps=0.24: 2400_vs_4800 | relative=0.004021% | abs_RMS=5.0421387e-07 | 1%=PASS | 0.1% target=PASS

N150 Python/C++ RMS=9.7e-18; serial-vs-parallel=0

N=1200 RK4 time: 4vs8=2.660e-10; 8vs16=1.662e-11; status=PASS

N=4800 RK4 time: 4vs8=2.667e-10; 8vs16=1.666e-11; status=PASS

B_Torus_2015_p66

Baseline 1%: PASS; temporal PASS; CPU Python/native N150 PASS

eps=0.06: 600_vs_1200 | relative=2.655875% | abs_RMS=0.00025307093 | 1%=FAIL | 0.1% target=FAIL

eps=0.06: 1200_vs_2400 | relative=0.679595% | abs_RMS=6.6049132e-05 | 1%=PASS | 0.1% target=FAIL

eps=0.06: 2400_vs_4800 | relative=0.169058% | abs_RMS=1.6515302e-05 | 1%=PASS | 0.1% target=FAIL

eps=0.12: 600_vs_1200 | relative=0.745904% | abs_RMS=6.5630603e-05 | 1%=PASS | 0.1% target=FAIL

eps=0.12: 1200_vs_2400 | relative=0.185547% | abs_RMS=1.6410836e-05 | 1%=PASS | 0.1% target=FAIL

eps=0.12: 2400_vs_4800 | relative=0.046327% | abs_RMS=4.1027192e-06 | 1%=PASS | 0.1% target=PASS

eps=0.24: 600_vs_1200 | relative=0.202633% | abs_RMS=1.6127256e-05 | 1%=PASS | 0.1% target=FAIL

eps=0.24: 1200_vs_2400 | relative=0.050595% | abs_RMS=4.0318531e-06 | 1%=PASS | 0.1% target=PASS

eps=0.24: 2400_vs_4800 | relative=0.012645% | abs_RMS=1.0079656e-06 | 1%=PASS | 0.1% target=PASS

N150 Python/C++ RMS=1.13e-17; serial-vs-parallel=0

N=1200 RK4 time: 4vs8=2.245e-13; 8vs16=6.510e-14; status=PASS

N=4800 RK4 time: 4vs8=2.364e-13; 8vs16=6.372e-14; status=PASS

4. Interpretation / remaining limitations

Convergence is observed for the specified discretized filament ODE at fixed epsilon, not for unregularized Euler or a finite-core fluid PDE.

Changing epsilon and N are separate operations. In particular, B at epsilon=0.06 fails 600 to 1200 at the 1 percent criterion, then improves on finer grids.

The two geometries are different historical formulas; their unlike total arclengths are NOT rescaled to an equal physical core size.

The same analytic curve sampled at N600 and N4800 does not count as multiple independent physical providers or multiple experimental measurements.

Future tests: self-confinement and source-native finite-core dynamical Euler pressure Poisson, vortex clock frequency, boundary response, held-out independent observations.

5. Provenance and references

G2/G11/G13 historical negative evidence remains; all true physical closure claims unresolved.

Saffman (1992), Vortex Dynamics, Cambridge University Press, doi:10.1017/CBO9780511624063.

Butcher (2008), Numerical Methods for Ordinary Differential Equations, Wiley, doi:10.1002/9780470753767.